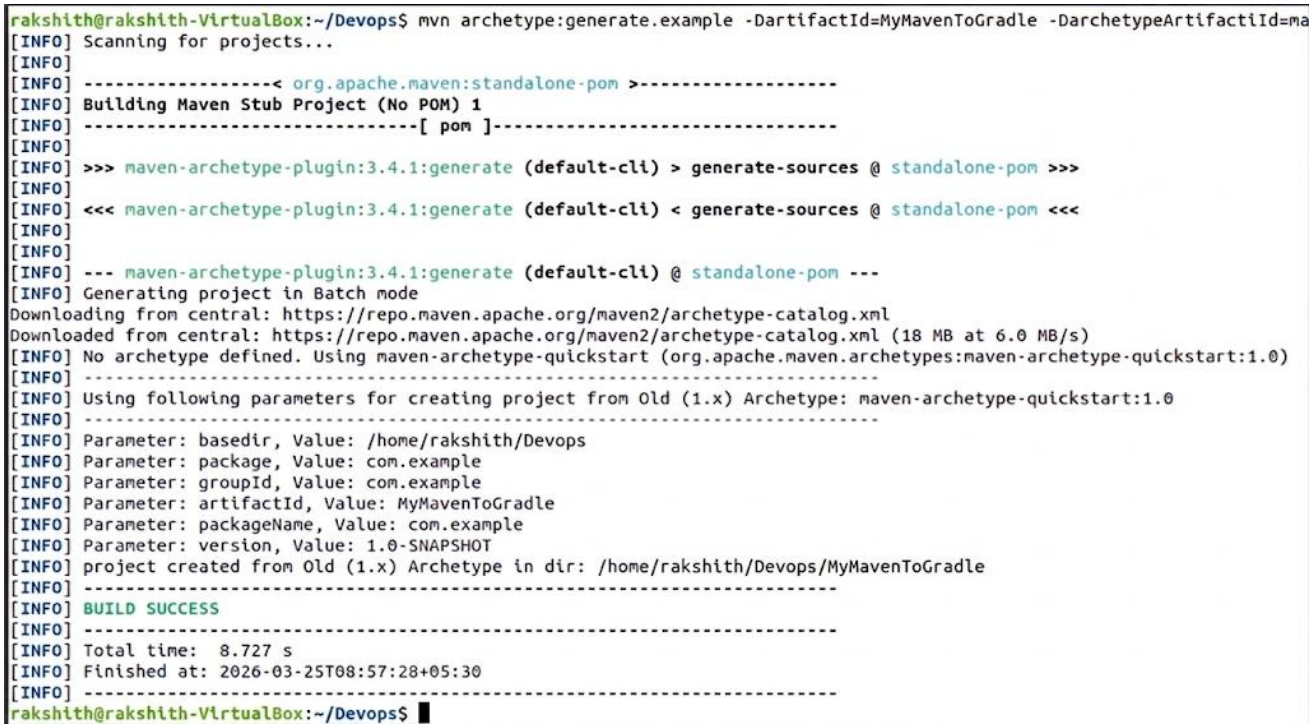
Program 9: Maven To Gradle

Step 1: Create a new maven project using this command

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenToGradle -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false



Step 2: Edit Pom.xml

<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example1</groupId>
<artifactId>MyMavenToGradle</artifactId>
<packaging>jar</packaging>
<version>1.0-SNAPSHOT</version>
<name>MyMavenToGradle</name>
<url>http://maven.apache.org</url>
<!-- Properties: Customize Java version or plugin versions -->
<properties>
<maven.compiler.source>11</maven.compiler.source>
<maven.compiler.target>11</maven.compiler.target>

```
</properties>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
</dependencies>
<!--  Build: Configuring plugins and build settings  -->
<build>
<plugins>
<!--  Example: Maven Compiler Plugin to compile Java code  -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.6</source>
<target>1.6</target>
</configuration>
</plugin>
<!--  Example: Maven Surefire Plugin to run tests  -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.1.0</version>
<configuration>
<archive>
<manifestEntries>
<Main-Class>com.example1.App</Main-Class>
</manifestEntries>
</archive>
</configuration>
</plugin>
</plugins>
</build>
</project
```

Step 3: Execute the command mvn clean install:

```
rakshith@rakshith-VirtualBox:~/Devops/MyMavenToGradle$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example1:MyMavenToGradle >--------------------
[INFO] Building MyMavenToGradle 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenToGradle ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenToGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/rakshith/Devops/MyMavenToGradle/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenToGradle ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/rakshith/Devops/MyMavenToGradle/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenToGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/rakshith/Devops/MyMavenToGradle/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenToGradle ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, 1.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/rakshith/Devops/MyMavenToGradle/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2.test (default-test) @ MyMavenToGradle ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.014 s - in com.example.AppTest
[INFO]
```

Step 4: When the command ‘gradle init -type pom.xml’ is executed, we get a error

```
rakshith@rakshith-VirtualBox:~/Devops/MyMavenToGradle$ gradle init --type pom.xml

FAILURE: Build failed with an exception.

* What went wrong:
Execution failed for task ':init'.
> The requested build setup type 'pom.xml' is not supported. Supported: 'basic', 'groovy-application', '

* Try:
Run with --stacktrace option to get the stack trace. Run with --info or --debug option to get more log o

* Get more help at https://help.gradle.org

BUILD FAILED in 6s
2 actionable tasks: 2 executed
rakshith@rakshith-VirtualBox:~/Devops/MyMavenToGradle$
```

Step 5: execute ‘gradle init -type pom’ to get build success

```
rakshith@Ubuntu:~/Devops/MyMavenToGradle$ gradle init --type pom

> Task :init
Maven to Gradle conversion is an incubating feature.


BUILD SUCCESSFUL in 5s
2 actionable tasks: 1 executed, 1 up-to-date
rakshith@Ubuntu:~/Devops/MyMavenToGradle$
```

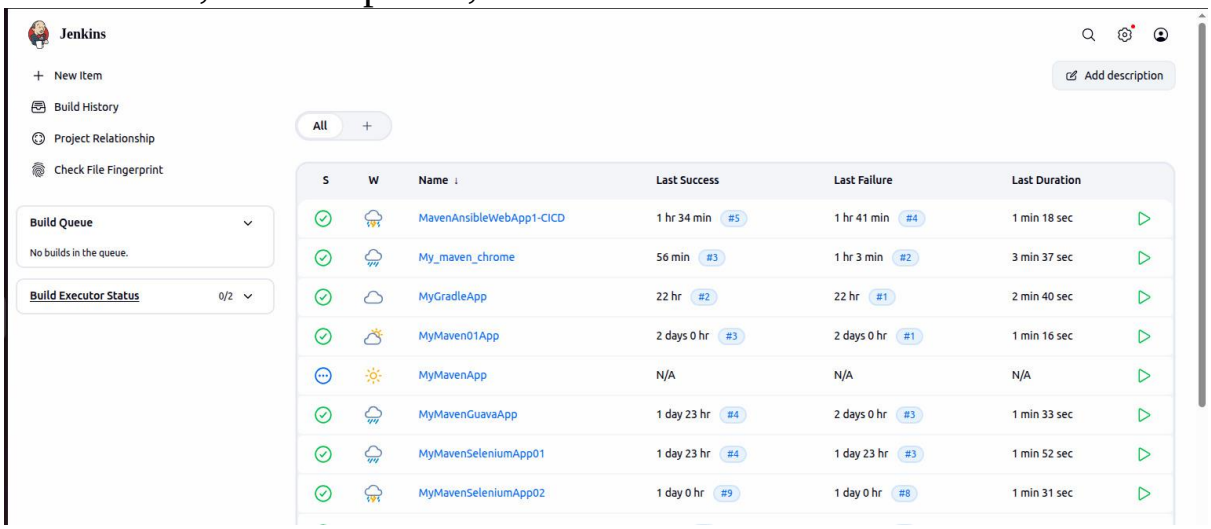
Step 6: Display tree structure

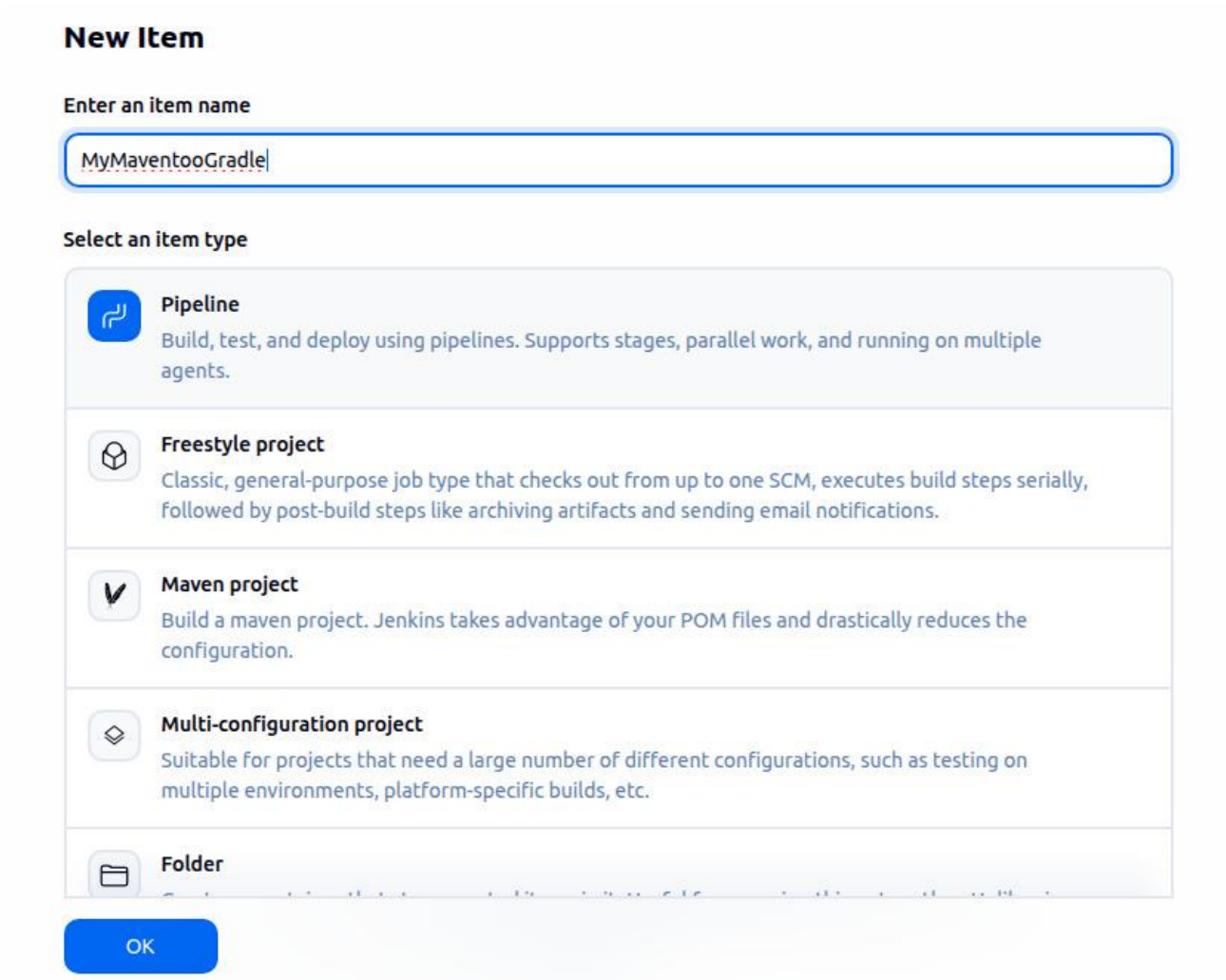
```
rakshith@rakshith-VirtualBox:~/Devops/MyMavenToGradle$ tree
.
├── build.gradle
├── gradle
│   └── wrapper
│       ├── gradle-wrapper.jar
│       └── gradle-wrapper.properties
├── gradlew
├── gradlew.bat
├── pom.xml
├── settings.gradle
├── src
│   ├── main
│   │   └── java
│   │       └── com
│   │           └── example
│   │               └── App.java
│   └── test
│       └── java
│           └── com
│               └── example
│                   └── AppTest.java
└── target
    ├── classes
    │   └── com
    │       └── example
    │           └── App.class
    ├── generated-sources
    │   └── annotations
    ├── generated-test-sources
    │   └── test-annotations
    ├── maven-archiver
    │   └── pom.properties
    ├── maven-status
    │   └── maven-compiler-plugin
    │       ├── compile
    │       │   └── default-compile
    │       │       ├── createdFiles.lst
    │       │       └── inputFiles.lst
    │       └── testCompile
    │           └── default-testCompile
    │               ├── createdFiles.lst
    │               └── inputFiles.lst
    ├── MyMavenToGradle-1.0-SNAPSHOT.jar
    ├── surefire-reports
    │   ├── com.example.AppTest.txt
    │   └── TEST-com.example.AppTest.xml
    └── test-classes
        └── com
            └── example
                └── AppTest.class
```

Jenkins Pipeline for MaventoGradle

Step 1: Create new Pipeline:

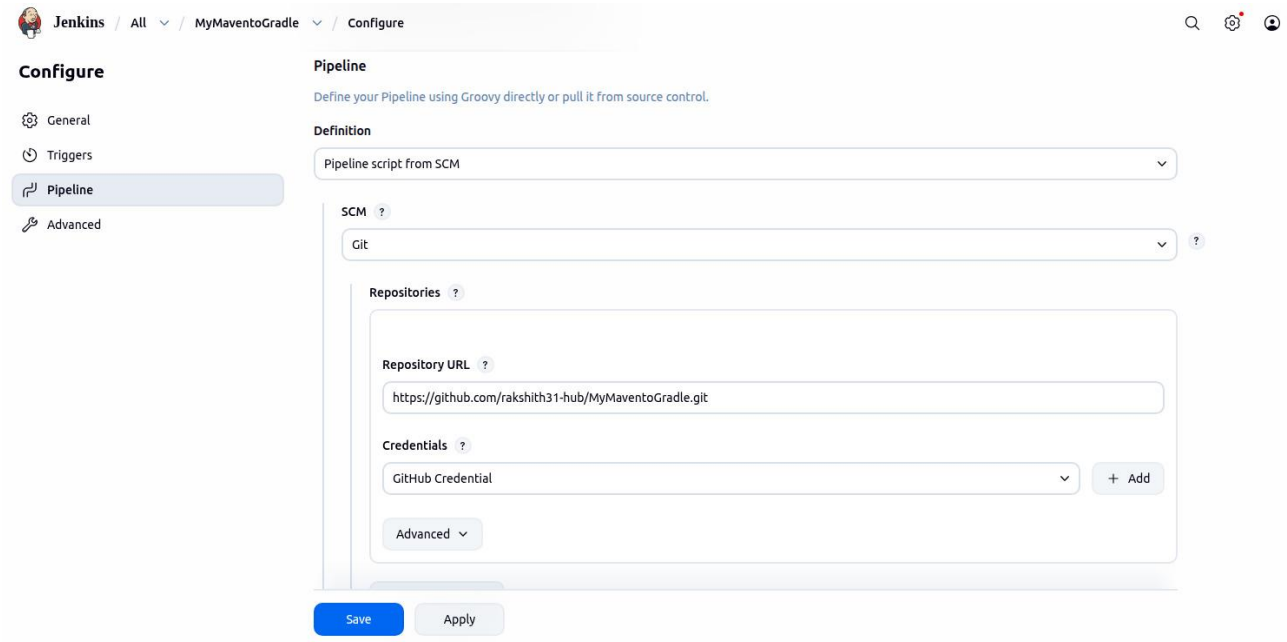
Go to Dashboard → + New Item.
Enter name, select Pipeline, and click OK.





Step 2: Configure Pipeline SCM:

Scroll down and select Pipeline script from SCM.
SCM: Git Enter repository URL and select GitHub credentials.

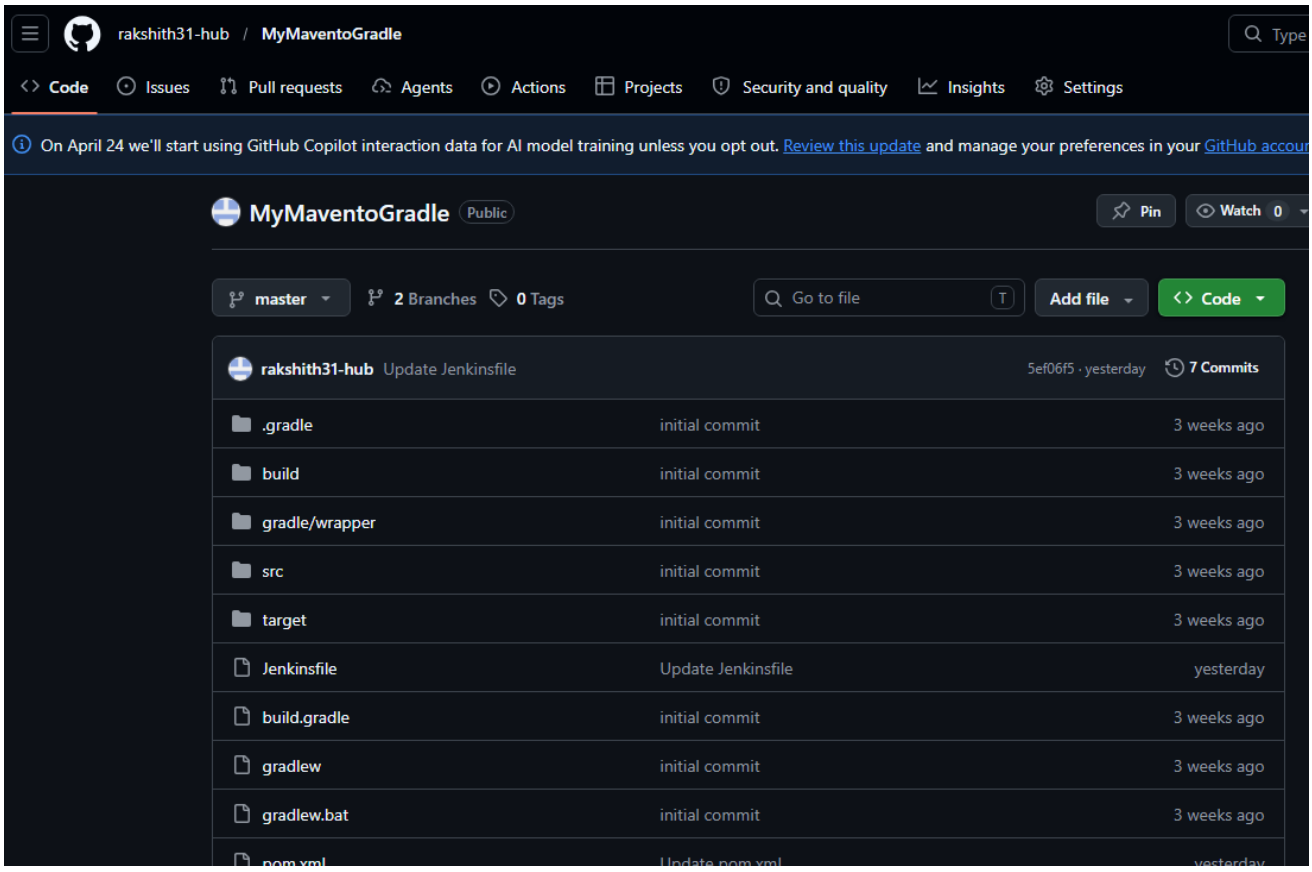


Step 3: Create Jenkinsfile

```
pipeline {
    agent any

    tools {
        maven 'Maven'
    }

    stages {

        stage('Checkout') {
            steps {
                git branch: 'main', url:
'https://github.com/rakshith31-
hub/MyMavenToGradle.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean package'
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'
            }
        }

        stage('Check Target Folder') {
            steps {
                sh 'ls -l target'
            }
        }

        stage('Run Application') {
            steps {
                sh 'java -jar target/MyMavenToGradle-1.0-SNAPSHOT.jar'
```

```
            }
        }

    }

    post {
        success {
            echo 'Build and execution successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 4: Now Push changes and commit by pushing the necessary “Jenkinsfile” to the respective Repo



Step 5: Build a successful Pipeline
Go to Jenkins and click Build Now.
Monitor Build Stages Use Stage View and Full Stage View to monitor execution.